

Overview

This report provides a standalone deep-dive for each of the three shortlisted CV+DL projects — remote photoplethysmography via RhythmMamba, deepfake/AI-image detection via MLEP+LOTA fusion, and silent lip reading via VALLR — together with a concrete two-person work split for each, so the group can execute whichever project is finally chosen without re-planning from scratch.

Project 1: RhythmMamba-Based Remote Heart Rate Sensing Under Gaming Stress

Technical Foundation

RhythmMamba is a state-space-model architecture for remote photoplethysmography (rPPG) — recovering a person's heart rate purely from facial video, with no contact sensor. It replaces the transformer backbones used in prior rPPG work with a Mamba-based sequence model, using a frame stem to extract spatial features, multi-temporal Mamba blocks to model short- and long-range temporal dependencies simultaneously, and a frequency-domain feed-forward block to sharpen the periodic heart-rate signal. It reports state-of-the-art accuracy on UBFC-rPPG, PURE, and MMPD benchmarks while running with 319% higher throughput and 23% lower peak GPU memory than the transformer baselines it replaces, making it realistically trainable on a single consumer GPU. The GitHub reference implementation confirms training/testing splits, checkpoint structure, and preprocessing pipeline are openly available, which de-risks reproduction for a two-person student team.^{[1][2][3][4]}

The Compulsory Delta (What Makes It Not a Copy)

The paper only validates on curated, low-motion lab datasets. Your project reframes the same architecture as a **gaming-stress heart-rate monitor**: capturing webcam video of yourselves during live Valorant/CS2 rounds (RGB keyboard glow, screen flicker, head movement, clutch-round facial strain) and measuring how far RhythmMamba's error degrades outside lab conditions, then patching the frequency feed-forward block with an attention-gated variant to partially recover accuracy — an ablation the original authors never test in this exact configuration.

CV Component (50%)

- Face detection and ROI (region-of-interest) tracking per frame using a lightweight detector (e.g., MediaPipe/RetinaFace) to isolate skin regions robust to head motion.
- Signal preprocessing: RGB channel decomposition, temporal normalization, motion-artifact rejection heuristics for the self-recorded gaming clips.
- Frame stem feature extraction pipeline (spatial feature maps feeding into the Mamba blocks).

DL Component (50%)

- Multi-temporal Mamba block implementation/reuse and training loop on UBFC-rPPG/PURE for baseline reproduction.
- Frequency-domain feed-forward block modification: swapping in an attention-gated frequency filter and re-training.
- Evaluation pipeline: MAE, RMSE, Pearson correlation between predicted and ground-truth heart rate, benchmarked lab-data vs. gaming-data.

Two-Person Task Split

Person A — CV & Data Lead	Person B — DL & Evaluation Lead
Set up face detection/ROI tracking pipeline and preprocessing (skin-region isolation, normalization)	Reproduce RhythmMamba baseline training on UBFC-rPPG/PURE using the public repo ^[^4]
Record and curate the self-shot gaming-session dataset (both members' faces during CS2/Valorant play) with synced ground-truth pulse (phone PPG app or fitness tracker)	Implement and train the attention-gated frequency feed-forward block variant
Build the motion-artifact/lighting-robustness preprocessing for the noisy gaming footage	Run comparative evaluation (lab vs. gaming domain) — MAE/RMSE/Pearson correlation tables and plots
Write the CV methodology section of the report/paper (ROI extraction, preprocessing, dataset construction)	Write the DL methodology section (architecture, training regime, ablation results) and lead the live demo integration

Project 2: MLEP + LOTA Fusion for Cross-Generator Deepfake/AI-Image Detection

Technical Foundation

This project fuses two independently published 2025 detection papers that were never combined by their original authors: one branch exploits multi-level entropy patterns (texture/statistical irregularities introduced by generative models), and the other branch exploits low-level bit-plane/compression artifacts (LOTA-style) left behind by GAN or diffusion pipelines. Because no existing paper performs this fusion, the fusion gate itself — learning when to trust texture-entropy evidence versus compression-artifact evidence — is your primary, defensible novelty claim, distinguishing this from simply re-running either paper's code.^[^5]

The Compulsory Delta

The base pairing is DL-heavy with a thin CV footprint, so a CV-side semantic region-of-interest (face/object segmentation) preprocessing stage is added before both branches, restoring a genuine 50/50 split. The fusion model is then stress-tested zero-shot on 2025-era generators (Stable Diffusion 3, FLUX, or Midjourney v6 samples) after training only on older ProGAN-style data — a generalization gap neither source paper reports — and a Grad-CAM/saliency layer is added so the two branches' disagreements are visually interpretable rather than a black box.

CV Component (50%)

- Semantic ROI detection/segmentation stage (face parsing or object segmentation) run before both detection branches to standardize input regions.
- Bit-plane/compression-artifact extraction pipeline (JPEG-domain or spatial-domain artifact maps) for the LOTA-style branch.
- Dataset curation and preprocessing across generator families (ProGAN training set + newly scraped/generated SD3/FLUX/Midjourney test set), including resolution/format normalization.

DL Component (50%)

- Multi-level entropy feature branch implementation and training.
- Fusion gating network design (learned weighting between the two branches) and joint training/fine-tuning.
- Grad-CAM/saliency visualization module and the cross-generator generalization evaluation (accuracy, AUC, per-generator breakdown).

Two-Person Task Split

Person A — CV & Data Lead	Person B — DL & Fusion Lead
Build the semantic ROI/segmentation preprocessing stage applied to all input images	Implement and train the multi-level entropy detection branch
Implement the bit-plane/compression-artifact extraction pipeline for the LOTA-style branch	Design and train the fusion gating network combining both branches
Curate and preprocess the cross-generator dataset (ProGAN training set + SD3/FLUX/Midjourney test set), handling licensing/collection	Build the Grad-CAM/saliency explainability layer and run the zero-shot cross-generator generalization tests
Write the CV methodology section (segmentation, artifact extraction, dataset construction)	Write the DL/fusion methodology section and lead the interpretability results discussion

Project 3: Personalized Silent Lip Reading with VALLR for Real-Time Esports Callouts

Technical Foundation

VALLR is a two-stage visual speech recognition system: a Vision Transformer encoder with a CTC head first converts cropped lip-region video into phoneme sequences, and a LoRA-fine-tuned large language model then reconstructs full sentences from those phonemes, achieving state-of-the-art word-error-rate on the LRS3 benchmark while needing only about 0.6% of the labeled training data required by comparable end-to-end systems. The authors explicitly flag persistent confusion between visually similar phonemes (e.g., "P" vs. "B") and homophones as an open limitation, since the model is trained and tested only on generic public-speaker datasets.^{[6][7][^8]}

The Compulsory Delta

Rather than reproducing generic LRS2/LRS3 numbers, the project personalizes VALLR via LoRA fine-tuning (its own stated efficient-adaptation mechanism) on a small, self-recorded, closed-vocabulary dataset of both members mouthing gaming callouts ("rotate," "smoke mid," "eco round," "clutch"), directly measuring the WER improvement from personalization. A lightweight disambiguation module targets the exact P/B and homophone confusion cases the paper leaves unresolved, and the final deliverable is packaged as a live, mic-muted "silent shot-caller" demo — an applied use case the paper never attempts, since VALLR is only evaluated offline on pre-recorded clips.

CV Component (50%)

- Lip-region detection and cropping pipeline (facial landmark detection → mouth ROI extraction) robust to webcam conditions.
- Self-recorded closed-vocabulary dataset creation: filming both members repeating the callout vocabulary under varied lighting/angle, with frame-level annotation.
- Real-time video capture and preprocessing pipeline for the live demo (frame buffering, ROI tracking at inference time).

DL Component (50%)

- ViT + CTC phoneme recognition stage: reproduce/fine-tune on the base VALLR checkpoint.
- LoRA personalization fine-tuning on the closed-vocabulary dataset, plus the phoneme-confusion disambiguation module (context re-ranker or viseme clustering fix).
- WER evaluation pipeline (generic pretrained vs. personalized model) and the LLM sentence-reconstruction integration for the live demo.

Two-Person Task Split

Person A — CV & Data Lead	Person B — DL & Personalization Lead
Build lip-region detection/cropping pipeline using facial landmarks	Set up and reproduce the base VALLR ViT+CTC phoneme recognizer from the public checkpoint
Film and annotate the closed-vocabulary esports-callout dataset for both members	Implement LoRA fine-tuning for personalization on the self-recorded dataset
Build the real-time capture/ROI-tracking pipeline for the live demo	Build the phoneme-confusion disambiguation module and run WER comparisons (generic vs. personalized)
Write the CV methodology section (lip detection, dataset construction, real-time pipeline)	Write the DL methodology section (LoRA setup, disambiguation module, WER results) and integrate the LLM sentence-reconstruction for the demo

Cross-Project Comparison for Final Selection

Criterion	rPPG (RhythmMamba)	MLEP+LOTA Fusion	Silent Lip Reading (VALLR)
Compute needed	Low — single consumer GPU sufficient[^1]	Medium — two branches + fusion training	Medium-High — ViT + LLM fine-tuning
Data collection burden	Medium — gaming footage + reference pulse device	Medium-High — cross-generator image collection	Medium — closed-vocabulary filming, smaller vocabulary keeps it manageable
Novelty defensibility	Strong (new domain + architecture ablation)	Strong (first-of-its-kind fusion)	Strong (personalization + resolves stated limitation)
"Uncommon and interesting" factor	High — biometric + gaming crossover, live pulse-during-clutch demo	Medium-High — timely (deepfakes/AI images), but many groups likely pick this theme	Very high — live silent-speech gaming demo is visually striking and rare
Live demo shareability	Strong (visible pulse spike during clutch)	Moderate (detection scores less viscerally engaging live)	Very strong (mute mic, text appears on screen)

Silent Lip Reading and rPPG both offer stronger live-demo spectacle for differentiating the presentation from typical deepfake-detection submissions, while MLEP+LOTA Fusion offers the most straightforward "first-of-its-kind" novelty claim on paper. All three retain a genuine 50/50 CV/DL split once the compulsory changes above are applied.

References

1. [cv-and-ddl-project-ideas-2.pdf](#)
2. [cv-and-dl-project-ideas.pdf](#)
3. [Created-on-14-July-2026-1-3.pdf](#)
4. [Medical Image Classification with KAN-Integrated ...](#) - by ON Manzari · 2025 · Cited by 55 — In this paper, we introduce the Medical Vision Transformer (Med...
5. [Medical Image Classification with KAN-Integrated Transformers ...](#) - The paper introduces MedViTV2, a novel medical image classification architecture that integrates KAN...
6. [논문 퀵 리뷰] [Rethinking Vision-Language Model in Face Forensics](#) - 이 CVPR 2025 논문에 관하여, 본 리뷰는 탐지 점수와 텍스트 설명을 생성하는 multi-modal 얼굴 위조 탐지기인 M2F2-Det을 요약합니다.
7. [Medical image classification with KAN-integrated ...](#) - by ON Manzari · 2025 · Cited by 52 — MedViTV2 enhances robustness for corrupted medical images in mu...
8. [论文审查] [Medical Image Classification with KAN-Integrated Transformers and Dilated Neighborhood Attention](#) - 该论文介绍了一种名为MedViTV2的新型医疗图像分类模型，结合了Kolmogorov-Arnold网络（KAN）和增强的扩张邻域注意力（DiNA），以解决现有模型在分类受损图片时的局限性。以下是对该...